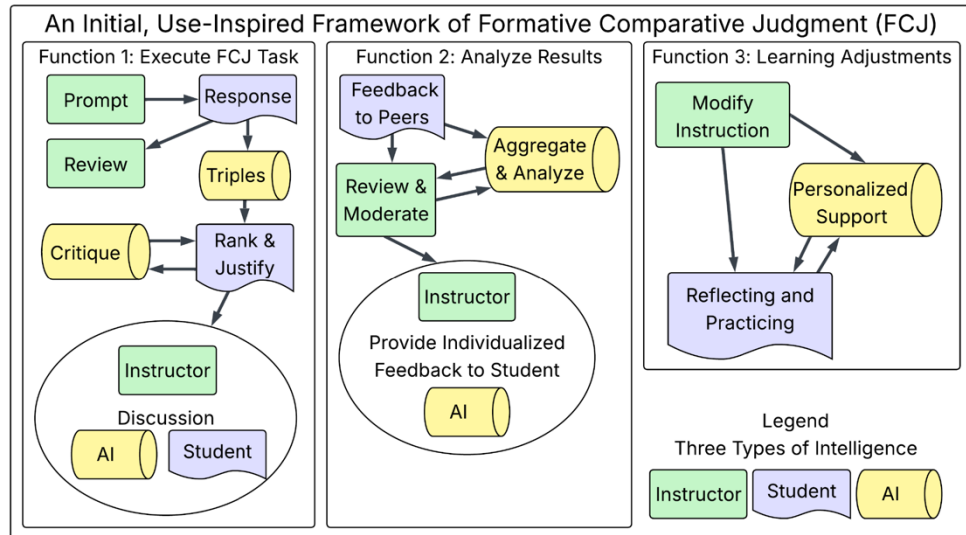


AI-support for teaching and learning at scale

- "write-to-learn" tasks improve learning outcomes, yet constructed-response methods become unwieldy when class sizes are large
- Logistical obstacles impair pedagogical best practices
- Investigating NLP tools with human-in-the-loop architecture to facilitate scalability—improve student benefit while mitigating instructor burden



Related Papers & Acknowledgement:

Beckman, Burke, Fiocchetta, Fry, Susan Lloyd, Patterson, Elle Tang (in review). Developing Consistency Among Undergraduate Graders Scoring Open-Ended Statistics Tasks. Preprint URL: <https://arxiv.org/abs/2410.18062>

Li, Susan Lloyd, Beckman, & Passonneau (2023). Answer-state Recurrent Relational Network (AsRRN) for Constructed Response Assessment and Feedback Grouping. *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP) 2023*

Susan Lloyd, Beckman, Pearl, Passonneau, Li, & Wang (2022). Foundations for AI-Assisted Formative Assessment Feedback for Short-Answer Tasks in Large-Enrollment Classes. In *Proceedings of the eleventh international conference on teaching statistics*. Rosario, Argentina.



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Matthew Beckman

Statistics & Data Science Education (SDSE) Research



<https://bit.ly/2026-Beckman-PSU>

mdb268@psu.edu
[mdbeckman.github.io](https://github.com/mdbeckman)



PennState

Other Projects/Interests

- Educational Assessment

Developing & analyzing tools to measure specific learning outcomes. Novel approaches to assessment.

"Tests" for research across academic institutions;

Formative assessment; e.g., feedback for self-regulated learning and/or adaptive instruction

> Sayali Phadke, Beckman, Lock Morgan (2024). Measuring contextualized statistical literacy: Evidence from an isomorphic assessment.

- Data Science Education

Intersection of stat education & computing education

Mixed-methods analysis of Intro DS Course content

Studying DS/CS topics introduced as learning objectives in statistics curricula

Data literacy, data acumen, EDA for Data Science

> Alyssa Hu, Hatfield, Beckman (2025). Exploring individuals' computational thinking with data. *ZDM Mathematics Education*, 57.

> Beckman et al (2021). Implementing Version Control With Git and GitHub as a Learning Objective in Statistics and Data Science Courses. *JSDSE*, 29(1).

Curious about SDSE?

- SDSE Research Lab (email me)
- Conferences: JSM; eCOTS 2026
- IASE catalog of PhD dissertations
- causeweb.org
- Talk to me!